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CUSTOMER SEGMENTATION WITH CLUSTERING

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EXECUTIVE SUMMARY

KEY FINDINGS AND CUSTOMER SEGMENTATION

Our analysis has identified three distinct customer segments based on purchase frequency and average unit cost:

1. Power Shoppers (High-Value Segment)
 - Highest purchase frequency and Customer Lifetime Value (CLV)
 - High average unit cost, but moderate to low recency (potential churn risk)
 - Represent the core of our business value
 - Recommendation: Implement targeted retention and reactivation campaigns
2. Regular Customers (Medium-Value Segment)
 - Moderate purchase frequency, CLV, and average unit cost
 - Stable engagement patterns
 - Recommendation: Create pathways to transition them to "Power Shoppers"
3. Occasional Shoppers (Low-Value, High-Recency Segment)
 - Lowest purchase frequency, CLV, and average unit cost
 - Most recent purchases, indicating active engagement
 - Recommendation: Develop programs to increase purchase frequency and value

PRIMARY VALUE DRIVERS AND RECOMMENDATIONS

- Purchase frequency and average unit cost strongly determine CLV
- Tailor strategies to increase both metrics for each segment

- Focus on retention for Power Shoppers, upselling for Regular Customers, and engagement for Occasional Shoppers

POTENTIAL BUSINESS IMPACT

By implementing segment-specific strategies, we anticipate:

- Revenue Growth: 15-20% increase by moving customers up the value chain
- Customer Retention: 10-15% improvement in "Power Shopper" retention rate
- CLV Enhancement: 25-30% increase for "Occasional Shoppers" through targeted engagement
- Marketing Efficiency: 20-25% improvement in ROI via segment-specific strategies

Visual References

Figure 1 in Appendix: CLV vs Frequency scatter plot, Distribution of Purchase Frequency and CLV vs Recency scatter plots

DATA PREPARATION AND QUALITY ASSURANCE

To ensure the highest quality of insights, we performed a thorough data preparation process. This included removing duplicate entries, addressing anomalies in customer age data, and ensuring each customer is represented consistently.

These steps allowed us to create a robust and reliable foundation for our analysis, giving us confidence in the accuracy of our customer segmentation.

CLUSTERING METHODOLOGY AND RATIONALE

To determine the optimal number of customer segments, we employed multiple clustering techniques and evaluation methods:

Elbow Method

- The elbow plot (Figure 4 in Appendix) shows the relationship between the number of clusters and the within-cluster sum of squares (WCSS).
- A clear "elbow" was observed at $k=3$, indicating diminishing returns for additional clusters beyond this point.

Silhouette Scores

- The elbow plot (Figure 4 in Appendix) shows the relationship between the number of clusters and the within-cluster sum of squares (WCSS).
- A clear "elbow" was observed at $k=3$, indicating diminishing returns for additional clusters beyond this point.

K-MEANS CLUSTERING

We compared k-means results for 3 and 4 clusters.

1. Centroid plots:
 - $K=3$: Shows clearer separation between clusters, especially along PC1.
 - $K=4$: Introduces an additional cluster, but there's significant overlap between clusters 1, 2, and 3.
2. Silhouette plots:
 - $K=3$: The average silhouette score appears slightly higher. Clusters 1 and 2 have consistently positive scores.
 - $K=4$: More variation in silhouette scores, with cluster 3 showing some negative values.
3. Cluster quality:
 - $K=3$: All clusters have a substantial number of data points and are more evenly sized.
 - $K=4$: The additional cluster (3) seems to split an existing cluster, potentially overfitting.
4. Interpretability:
 - $K=3$ offers a simpler, more interpretable model that still captures major patterns in the data.

HIERARCHICAL CLUSTERING

The dendrogram (Figure 6 in Appendix) supports a 3-cluster solution.

Three distinct branches are visible at a higher level of the hierarchy, aligning with our k-means results.

FINAL DECISION

We opted for the 3-cluster solution based on:

1. Clear elbow in the WCSS plot
2. Better interpretability and actionability of segments
3. Consistency across multiple clustering methods
4. Alignment with business objectives for targeted marketing strategies

This approach balances statistical rigour with practical applicability, providing a robust foundation for our customer segmentation strategy.

STRATEGIC RECOMMENDATIONS

TARGETED MARKETING STRATEGIES

1. Power Shoppers:
 - Exclusive early access to new products
 - VIP events and personalised styling services
 - Advanced loyalty tiers with unique rewards
2. Regular Customers:
 - Personalised bundle offers to increase average order value
 - Gamified loyalty program to boost purchase frequency
 - Educational content to deepen product engagement
3. Occasional Shoppers:
 - Re-engagement campaigns with limited-time offers

- Social proof marketing highlighting product value
- Easy-to-join loyalty program with immediate benefits

PRODUCT DEVELOPMENT INSIGHTS

- Introduce limited edition items for Power Shoppers
- Develop mid-range product lines with upgrade options for Regular Customers
- Create entry-level products with clear value propositions for Occasional Shoppers

CUSTOMER LIFECYCLE MANAGEMENT

- Implement predictive models to identify at-risk Power Shoppers
- Design clear pathways for segment transitions with targeted incentives
- Personalise post-purchase experiences to reinforce buying decisions

IMPLEMENTATION PLAN

SHORT-TERM ACTIONS (0-6 MONTHS)

- Launch segment-specific email and social media campaigns
- Implement A/B testing for personalised website experiences
- Develop and roll out tiered loyalty program
- Train customer service team on segment-specific approaches

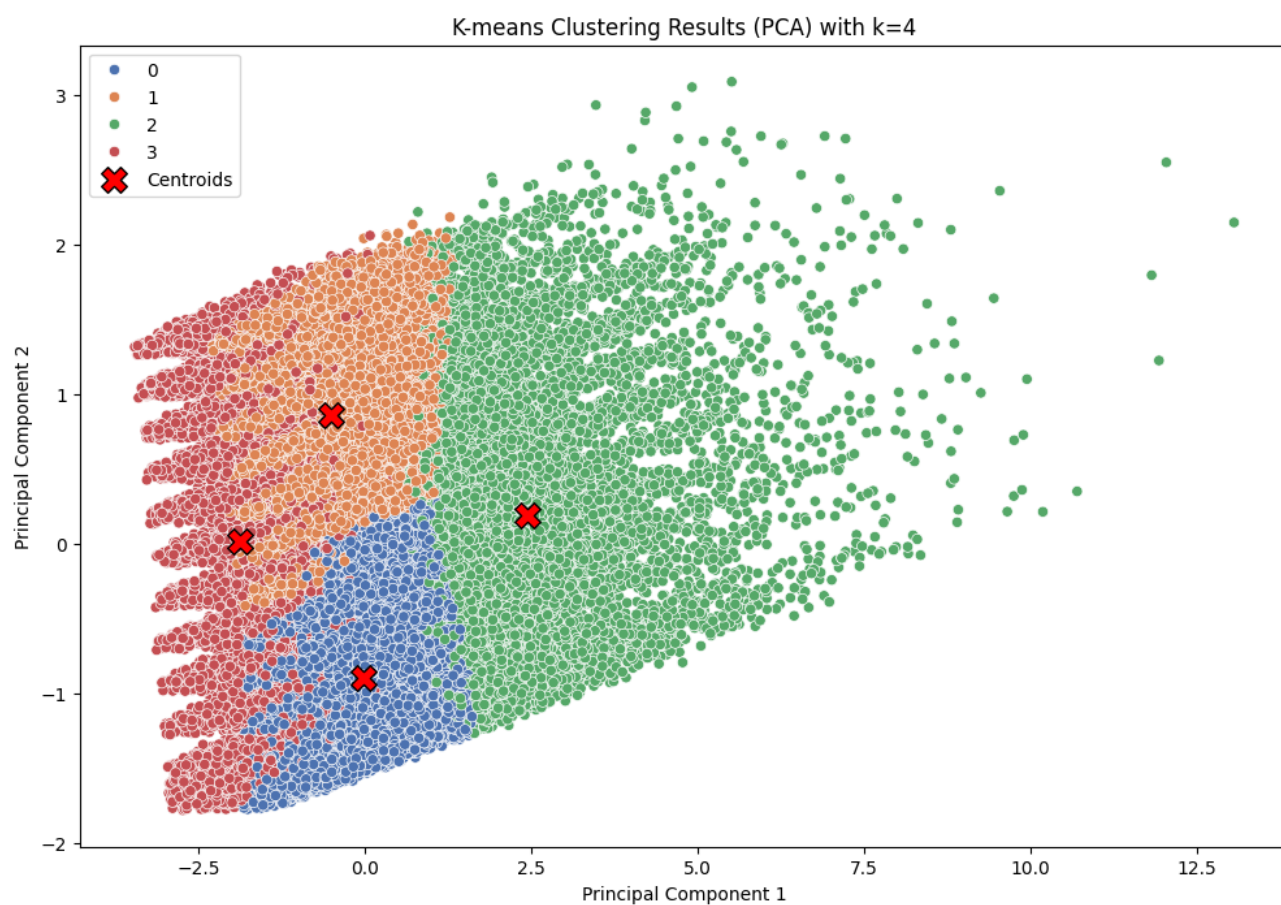
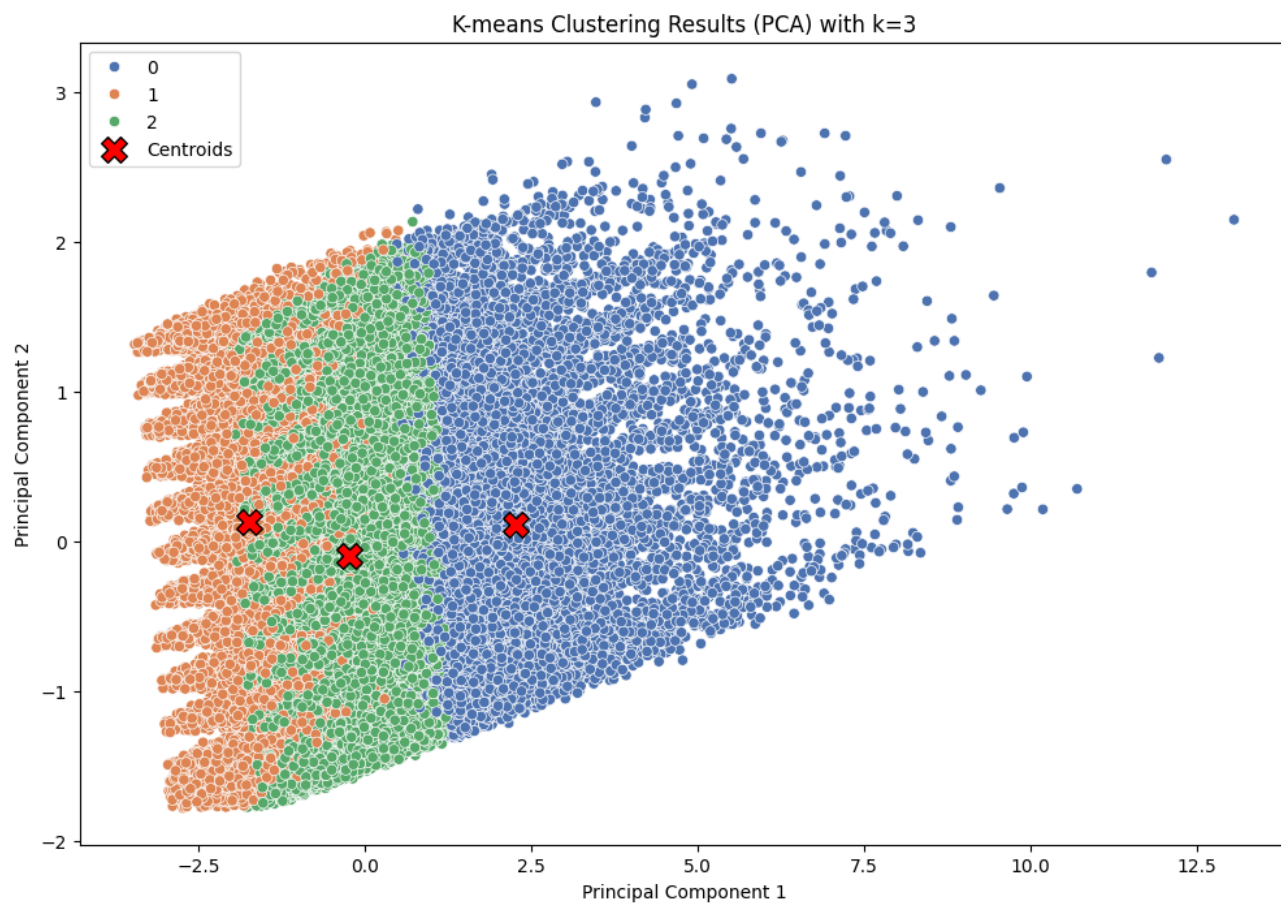
LONG-TERMS STRATEGIES (6-18 MONTHS)

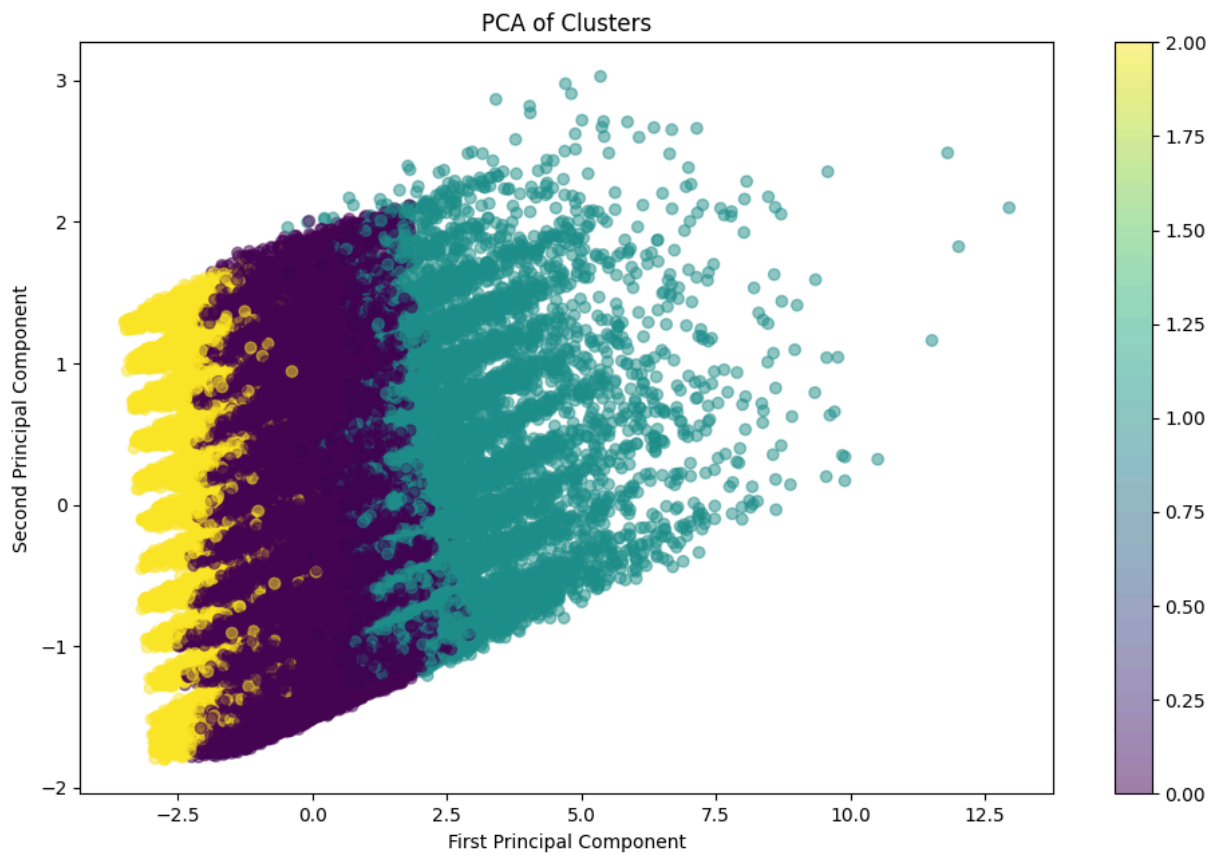
- Develop AI-driven product recommendation engine
- Create omni-channel customer journey maps for each segment
- Establish customer feedback loop for agile product development
- Implement predictive analytics for inventory management

KEY PERFORMANCE INDICATORS

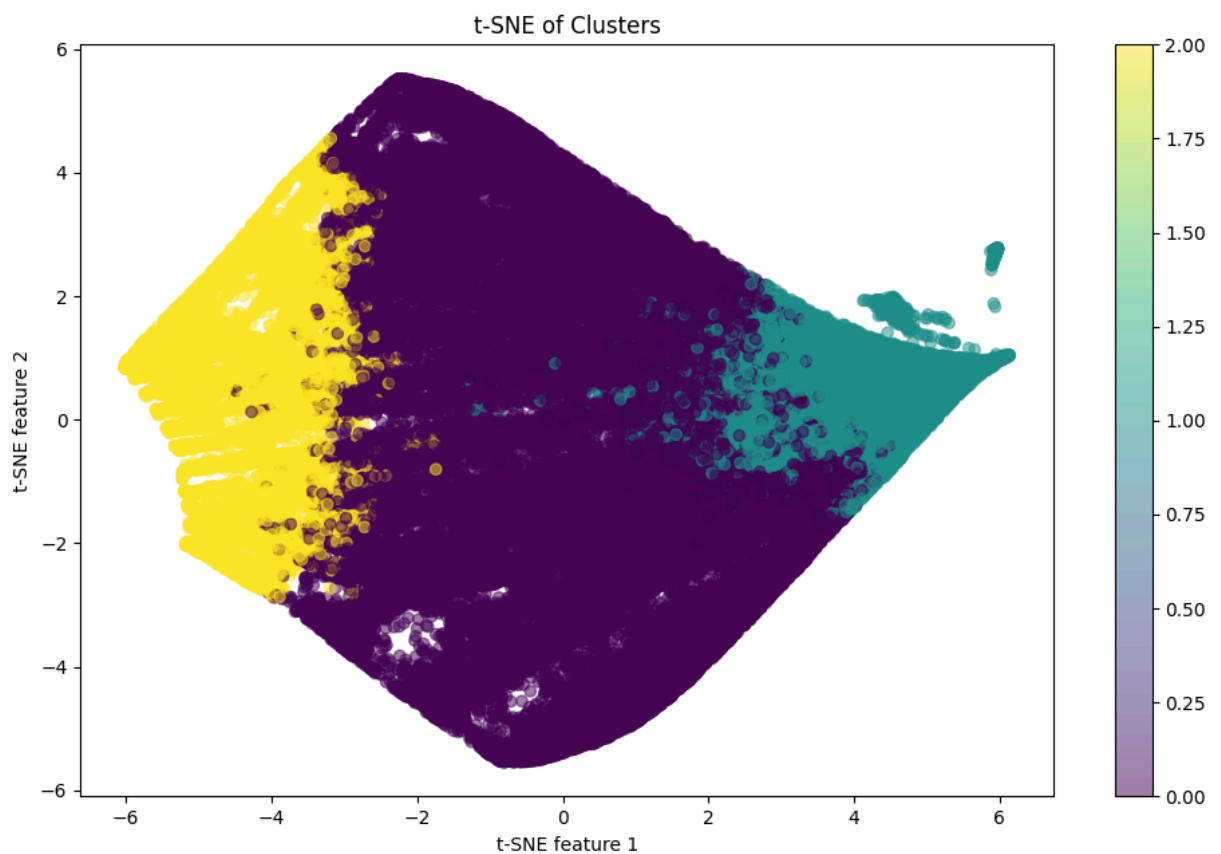
1. Segment Transition Rate (% of customers moving to higher-value segments)
2. Customer Lifetime Value (CLV) growth rate
3. Purchase Frequency increase by segment
4. Average Order Value (AOV) growth
5. Retention Rate by segment
6. Net Promoter Score (NPS) improvement
7. Return on Marketing Investment (ROMI) by segment

CUSTOMER SEGMENTATION WITH CLUSTERING

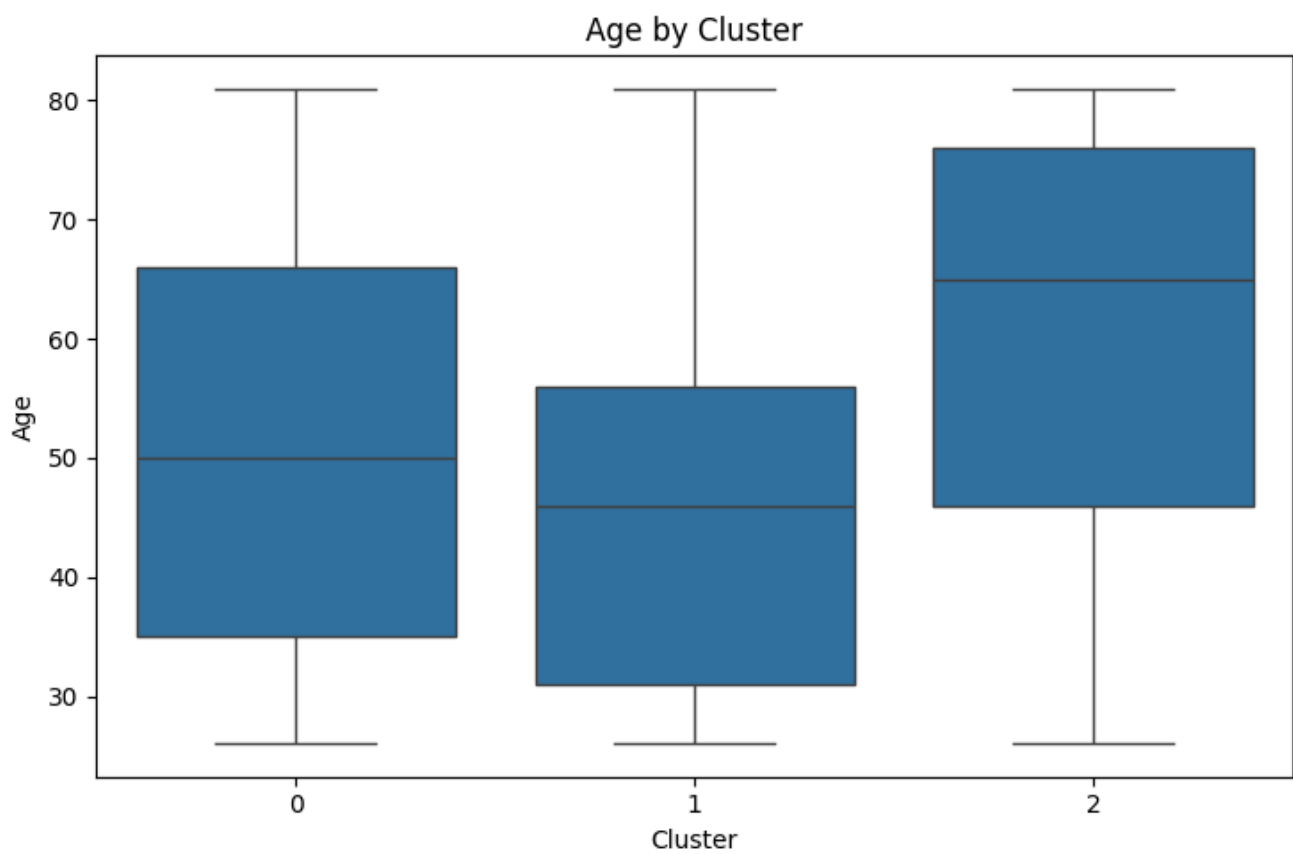
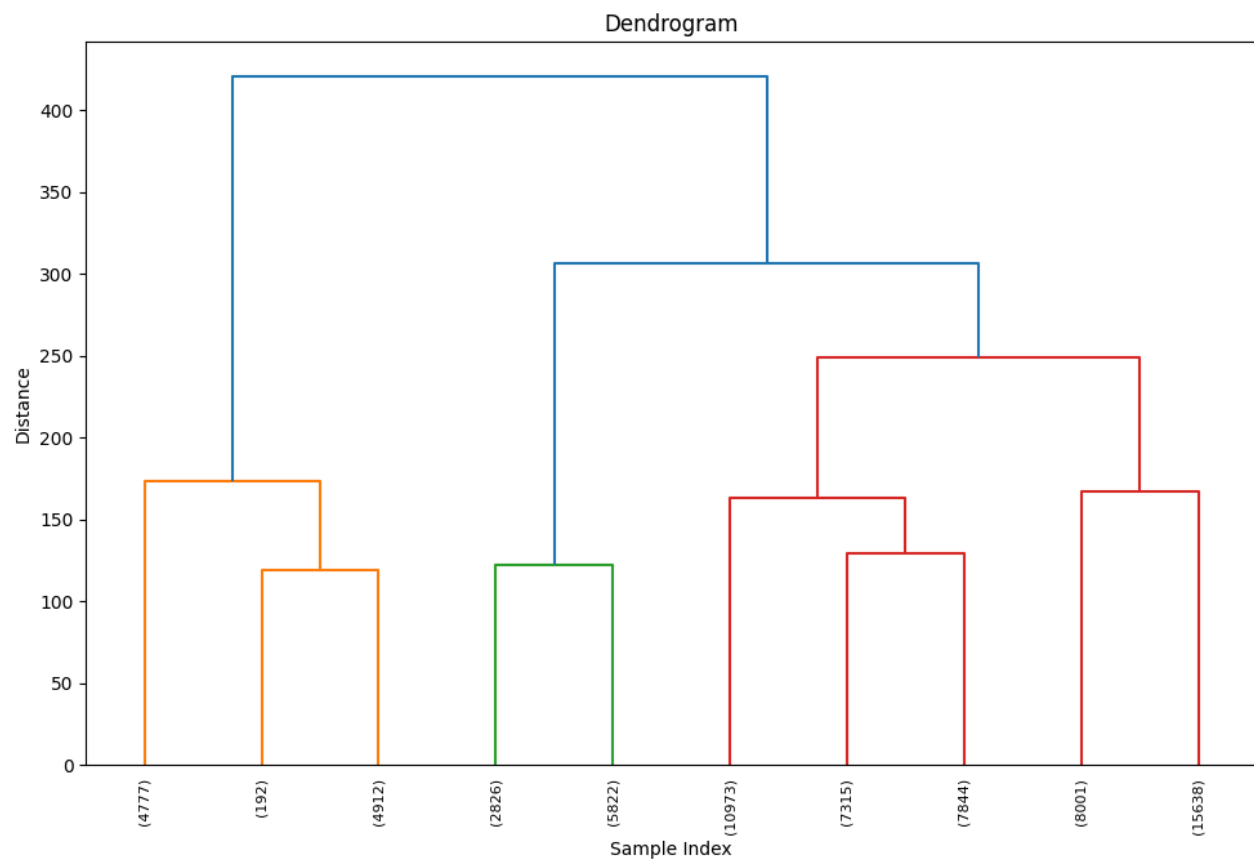




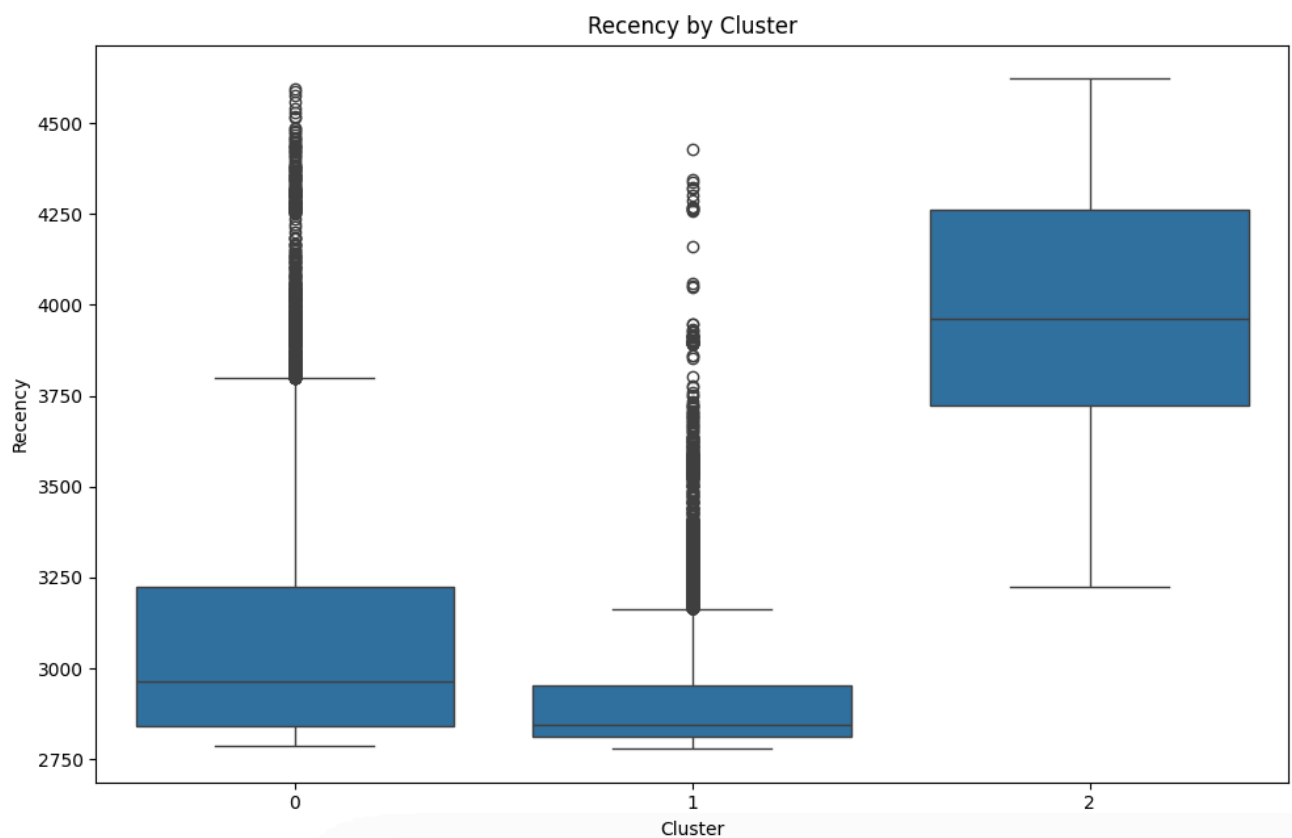
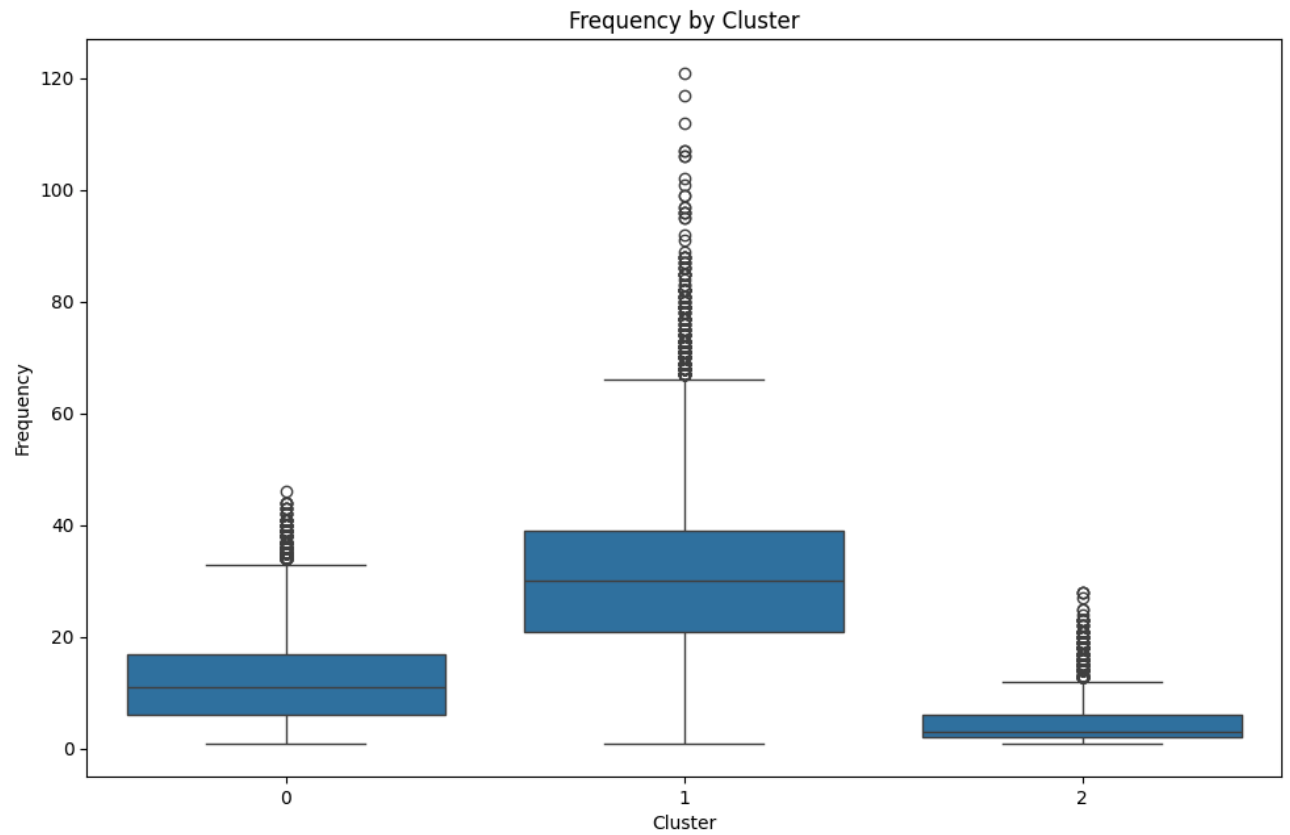
APPENDIX



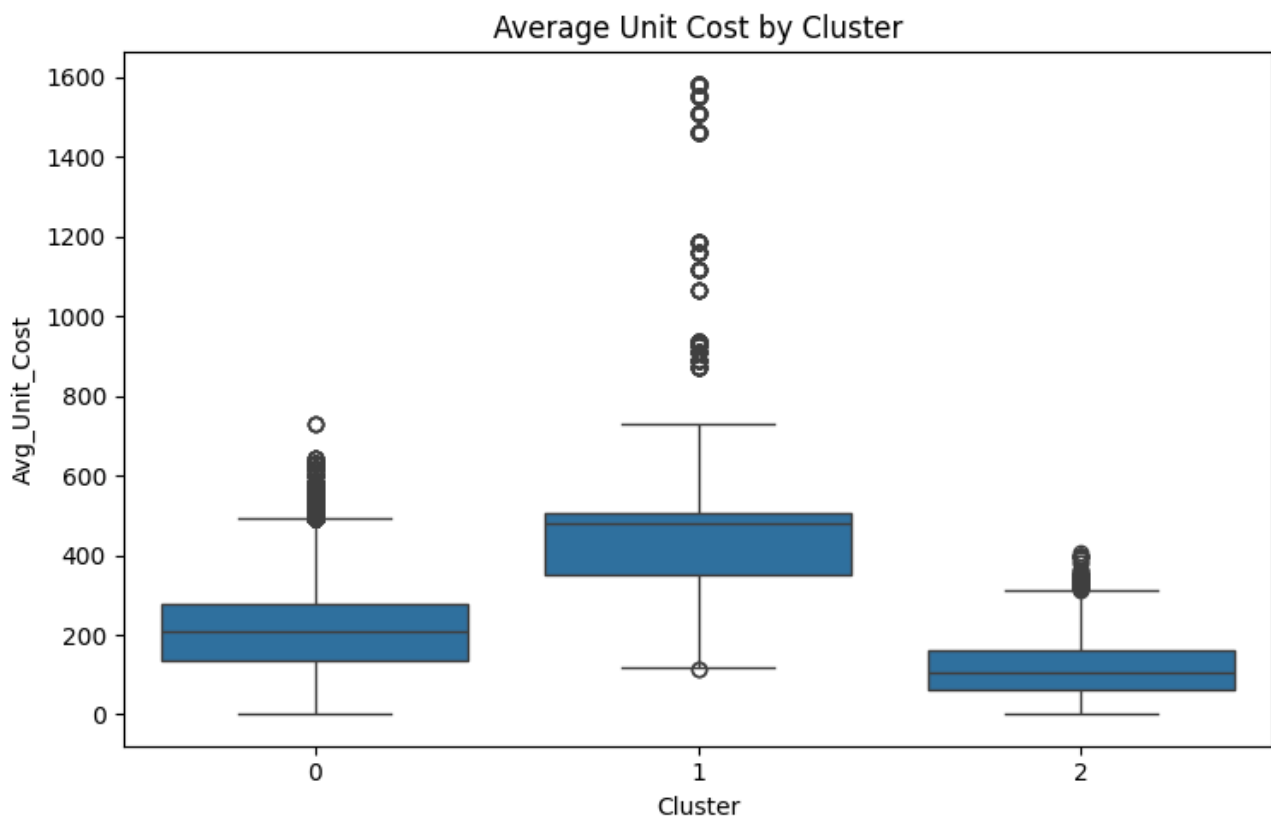
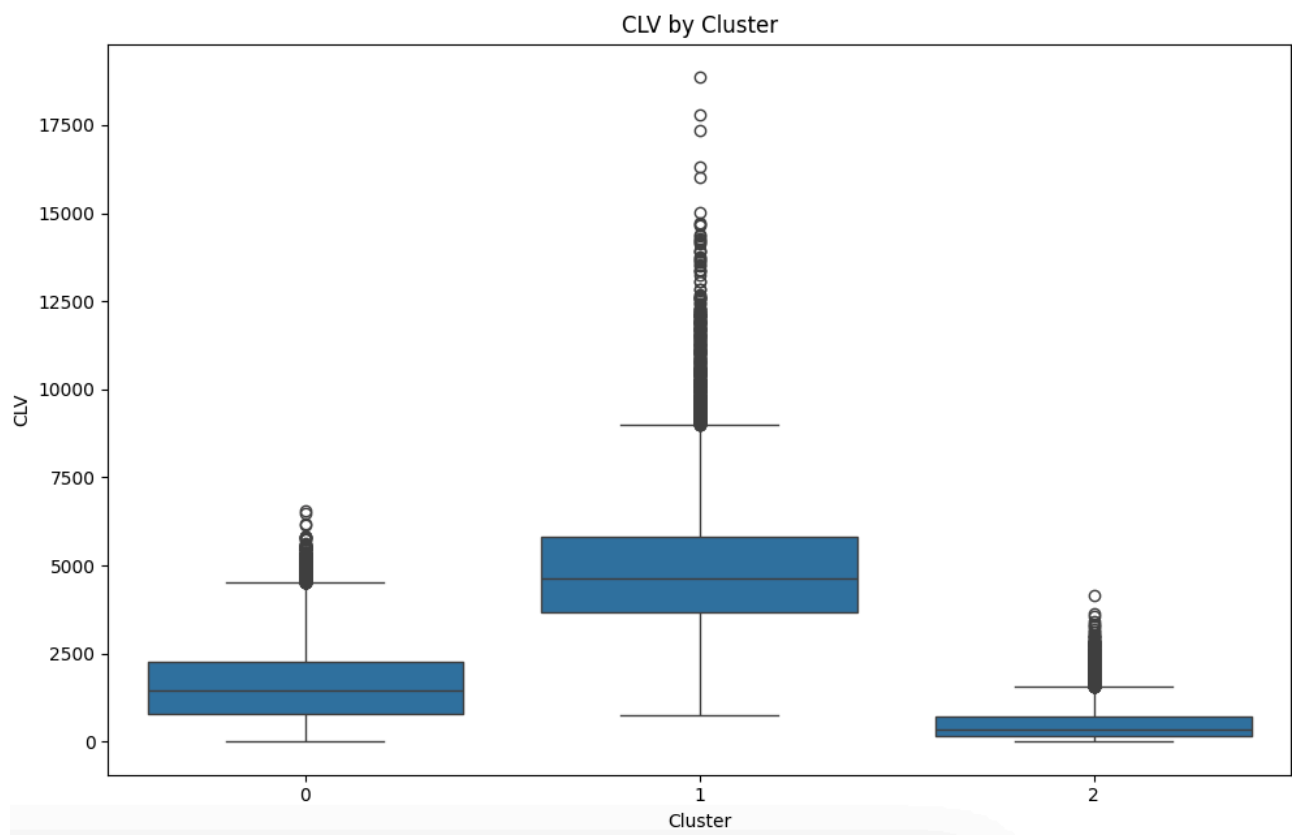
CUSTOMER SEGMENTATION WITH CLUSTERING



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DETAILED METHODOLOGY

FULL SET OF VISUALISATIONS

Figure 1

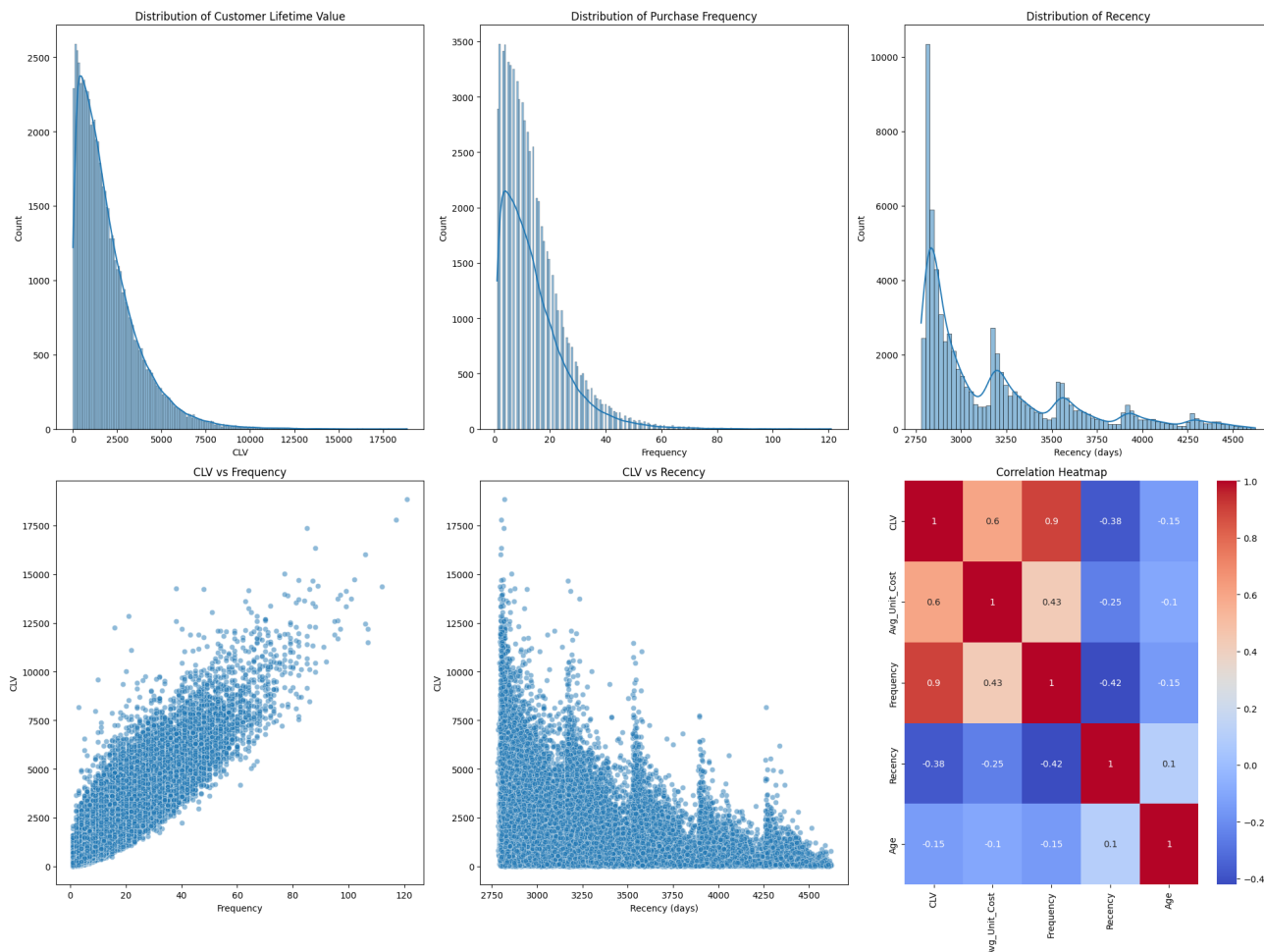


Figure 2

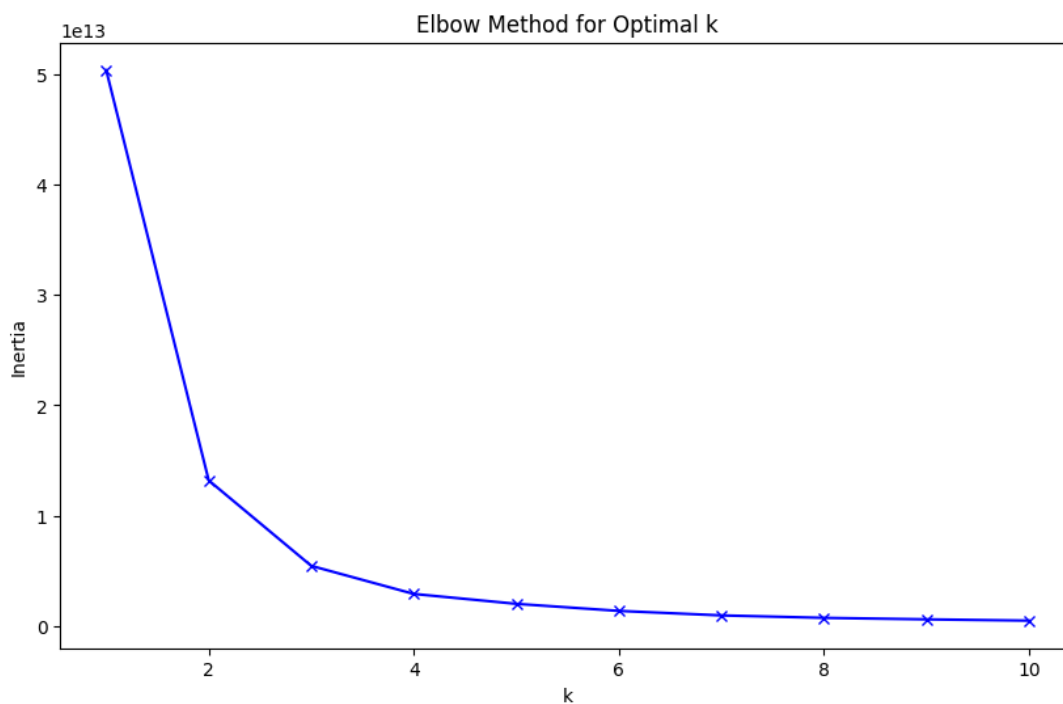


Figure 3

Figure 4

Figure 5

Figure 6

Figure 7

Figure 8

Figure 9

Figure 10

Figure 11

Figure 12